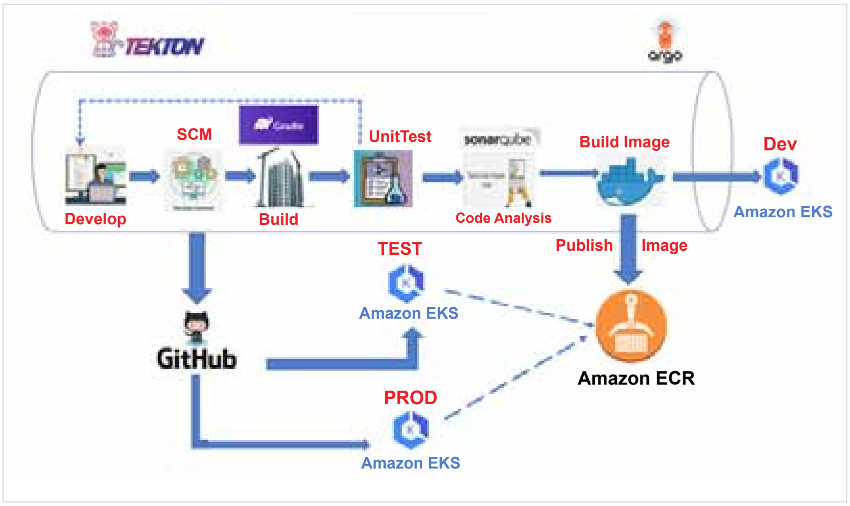

Figure 1: Tekton

Tekton provides several benefits to builders and users of CI/CD systems.

Customisable: Tekton entities are fully customisable, allowing for a high degree of flexibility. Platform engineers can define a highly detailed catalogue of building blocks for developers to use in a wide variety of scenarios.

Reusable: Tekton entities are fully portable; so once defined, anyone within the organisation can use a given pipeline and reuse its building blocks. This allows developers to quickly build complex pipelines without 'reinventing the wheel'.

Expandable: Tekton Catalog is a community-driven repository of Tekton building blocks. You can quickly create new and expand existing pipelines using pre-made components from the Tekton Catalog.

Standardised: Tekton installs and runs as an extension on your Kubernetes cluster and uses the well-established Kubernetes resource model. Tekton workloads execute inside Kubernetes containers.

Scalable: To increase your workload capacity, you can simply add nodes to your cluster. Tekton scales with your cluster without the need to redefine your resource allocations or any other modifications to your pipelines.

As a Kubernetes-native solution, Tekton seamlessly integrates with Kubernetes clusters. This allows developers to define and execute pipelines as Kubernetes Custom Resource Definitions (CRDs), making it easy to manage and monitor their pipelines alongside other Kubernetes resources.

Tekton pipelines embrace a modular approach to CI/CD, where tasks can be combined into customisable pipelines. This modularity fosters reusability and simplifies the maintenance of complex workflows. Teams can share and collaborate on tasks, promoting a culture of automation and efficiency.

These pipelines follow a declarative approach, allowing developers to specify what they want to achieve without being concerned with the underlying implementation details. This abstraction enhances pipeline maintainability and readability. Moreover, Tekton supports versioning, ensuring that pipelines remain consistent over time.

They leverage Kubernetes' scalability and isolation capabilities. Each task in a pipeline runs as a separate container, providing isolation and security. Teams can easily scale pipelines based on demand, ensuring optimal resource utilisation.

Tekton integrates well with a wide array of tools and platforms, including Kubernetes, Helm, Git, and more. This enables teams to create end-to-end delivery pipelines using their preferred tools while leveraging Tekton's core capabilities.

Tekton components

The Tekton ecosystem comprises various components that serve as the foundation for creating efficient and flexible CI/CD pipelines. These components seamlessly integrate within Kubernetes environments to orchestrate complex workflows with ease.

Tekton pipelines are the foundation of Tekton. They define a set of Kubernetes custom resources that act as building blocks from which you can assemble CI/CD pipelines.

Tekton triggers allow you to instantiate pipelines based on events. For example, we can trigger the instantiation and execution of a pipeline every time a PR is merged against a GitHub repository.

Tekton CLI provides a command-line interface called tkn, built on top of the Kubernetes CLI, which allows you to interact with Tekton.

Tekton dashboard is a web-based graphical interface for Tekton pipelines that displays information about the execution of your pipelines. It is currently a work-in-progress.

Tekton catalogue is a repository of high-quality, community-contributed Tekton building blocks — tasks, pipelines, and so on — that are ready for use in your own pipelines.

Tekton hub is a web-based graphical interface for accessing the Tekton Catalog.

Tekton operator is a Kubernetes operator pattern that allows you to install, update, and remove Tekton projects on your Kubernetes cluster.

Tekton chain provides tools to generate, store, and sign provenance for artifacts built with Tekton pipelines.

We can work with Tekton using tkn cli or Kubectl or Tekton API (available only for pipelines and triggers).